CLUSTERING:

Clustering

clustering:

clustering means dividing data points into homogeneous groups.

→ clustering is unsupervised : the data has no labels. The algorithm discovers the group by itself.

Example:

Grouping houses by value, type and location.

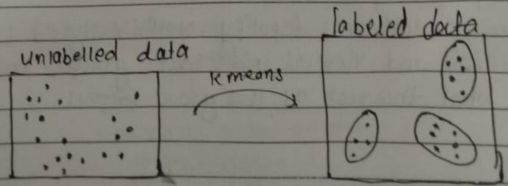
Grouping earthquake zones : area hit by past earthquakes cluster together, which helps predict where the next one is ~~pr~~ probable.

Application of clustering:

(1) Clustering helps in identification of groups of house on the basis of their value, type and geographical location.

2) clustering is used to study earth-quake. Based on the areas hit by an earthquake in a region, clustering can help analysis the next probable location where earthquake can occur.

Raw Data → Clustering Alg → CUSTERS OF DATA



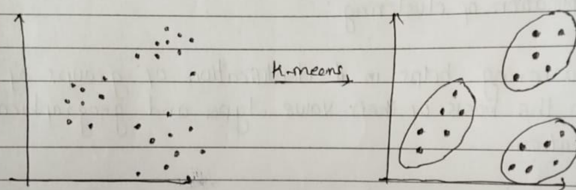

KMEANS:

# Kmeans

* About K Means clustering :

-> A k-means clustering algorithm tries to group Similar items in the form of clusters.
-> The no. of groups is represented by k. If k=2, there will be two cluster.
-> The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.

Before k-means




Q. Where is k-means used in Real Life Products ?

-> costumer Segmentation : e commerce and telecom companies cluster costumers by spending and behaviour, then target each segment differently. This is the no. one industry use of k-means
-> Streaming platforms (spotify, Netflix style) : cluster listeners and viewers into taste groups. then recommend What the rest of the group enjoys

→ News grouping (Google New style) : cluster similar articles on one story shows as a single group.

# k Means Algorithm steps :

Step 1 : select the no. of k to decide the no. of clusters.

Step 2 : select random k points or centroids.

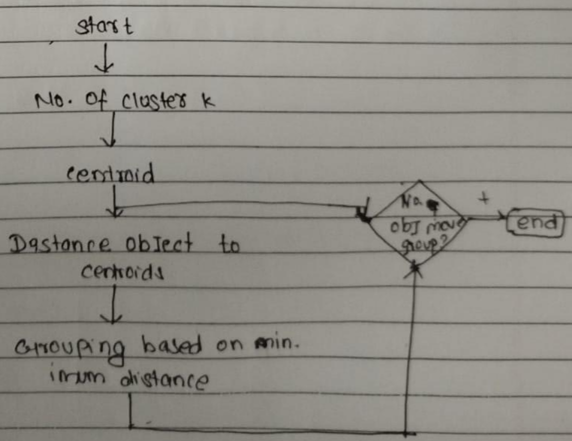
Step 3 : Assign each data points to their ~~clos~~ closest centroid, which will form the predefined k clusters.

Step 4 : calculate the variance and place a new centroid of each clusters.

Step 5 : Repeat the 3rd steps, which means ressign each data points to the new closest centroid of each cluster.

Step 6 : If any reassignment occurs, then go to step 4 else go to finish

Step 7 : The Model is ready.

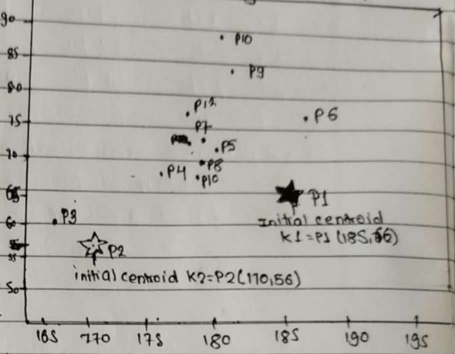


## Numerical

Data :- We cluster 12 people by height and weight with $k=2$

| Person | Height | weight |
|---|---|---|
| P1 | 185 | 65 |
| P2 | 170 | 56 |
| P3 | 168 | 60 |
| P4 | 179 | 68 |
| P5 | 182 | 72 |
| P6 | 188 | 77 |
| P7 | 180 | 71 |
| P8 | 180 | 70 |
| P9 | 183 | 84 |
| P10 | 180 | 88 |
| P11 | 180 | 67 |
| P12 | 177 | 76 |

Step 1: Pick $k=2$ & take P1, P2 as starting centroids.




- **Step 1** Point P3 (168, 60) :

We use Euclidean distance : $d = \sqrt{(x_2-x_1)^2 + (y_1-y_2)^2}$

Distance from P3 to k1 (185, 65)

$$d_{k1} = \sqrt{(168-185)^2 + (60-65)^2} = 17.72$$

Distance from P3 to k2 (170, 56)

$$d_{k2} = \sqrt{(168-170)^2 + (60-56)^2} = 4.472.$$

compare and assign.

→ Min is $d_{k2}$, 4.472, So P3 goes to the k2 group. $K_2 = (P_2,$

update centroid k2 (Mean of its points)

$$K_{2\,new} = \left(\frac{170+168}{2}, \frac{56+60}{2}\right) = (169, 58)$$

Now, $k_1 = (185, 65)$

$k_2 = (169, 58)$

- Step 2 : Point P4 (179, 68)

P4 to K1 :

$dk_1 = \sqrt{(179-185)^2 + (68-65)^2} = 6.70$

P4 to K2 : $dk_2 = \sqrt{(179-169)^2 + (68-58)^2} = 14.14$

→ Minimum is $dk_1 = 6.70$. So P4 goes to K1 group.

$K_{1new} = \left(\frac{185+179}{2}, \frac{68+65}{2}\right) = (182, 66.5)$

Now, $k_1 = (182, 66.5)$ and $k_2 = (169, 58)$.

- Step 3 and step 4 : Points P5 and P6.

~~Step 3 : P5 (185, 72) against K1 (181, 85)~~

~~$dk_1 = \sqrt{(182-182)^2 + (72-65)^2} =$~~

~~$dk_2 = \sqrt{18}$~~

- Step 3 : point P5 (182, 72)

P5 to $k_1 = \sqrt{(182-182)^2 + (72-66.5)^2} = 5.5$.

P5 to $k_2 = \sqrt{(182-169)^2 + (72-58)^2} = 19.10$.

→ Minimum is $dK_1 = 5.5$ so, P5 goes to K1 group.

$K_{1new} = \left(\frac{182+182}{2}, \frac{66.5+72}{2}\right) = (182, 69.25)$.

$k_1 = (182, 69.25)$, $k_2 = (169, 58)$

- Step 4 : The Remaing Points (Same Recipi)

for every points : distance to k1, distance to k2, take the minimum, assign, update that centroid

| step | Point | $dk_1$ | $dk_2$ | Minimum | goes to | Upd… |
|---|---|---|---|---|---|---|
| 4 | P6(188,77) | 9.80 | 26.87 | dk1 | k1 | k1 (185,7… |
| 5 | P7(180,71) | 5.43 | 17.02 | dk1 | k1 | (182.5, 72… |
| 6 | P8(180,70) | 2.71 | 16.27 | dk1 | k1 | (182, 72) |
| 7 | P9(183,84) | 2.23 | 29.32 | dk1 | k1 | (182, 73… |
| 8 | P10(180,88) | 14.77 | 31.95 | dk1 | k1 | (182, 75… |
| 9 | P11(180,67) | 8.52 | 14.21 | dk1 | k1 | (181.9, 7… |
| 10 | P12(177,76) | 5.17 | 19.70 | dk1 | k1 | (181.4, 74… |

Notice : K2 never ~~change~~ wins again after P3, so K2 stays at (169, 58) whole time. In one more pass over all the points, no point changes its cluster, so the algorithm stops.

final Result :

The two clusters :

k1 = {P1, P4, P5, P6, P7, P8, P9, P10, P11, P12}
k2 = {P2, P3}.

final centroids :

k1 = (181.4, 74.5)
k2 = (169.0, 58.0)

Real meaning : the algorithm discovered a tall and heavier group and a shorter and lighter group ~~and a shorter and~~ without any labels given to it.

Advantages and disadvantages :

- Advantages :
  - → Very simple to implement and explain.
  - → Scales to huge datasets and runs fast
  - → Adaps easily when new examples arrive
  - → works for many shapes and sizes of clusters.

- Disadvantages :
  - → Sensitive to outliers (they drag the centroid)
  - → choosing k manually is tough job.
  - → Result depends on the starting centroids
  - → Struggles when dimensions become very many.

Tips for choosing k :

Try several values of k and use the elbow method : plot the total distance for each k, and Pick k when curve bends like an elbow.

* Summary :

→ Clustering groups unlabeled data : Similar points togethers different point a part.

→ k-means is centroid based : every point belongs to its nearest centroid, and the goal is to minimize the point to centroid distance.

→ The recipe for each point : distance to k1, distance to k2, take the MINIMUM, assign the point to that group, and update that centroid to the mean of its points.

→ Real products use it everywhere : ~~clust~~ Costomer Segments, taste groups on streaming apps, fraud detection.